DATA ANALYTICS

NOTES / DVI

WHAT'S INSIDE

12 Core Topics • Diagrams & Flowcharts • Comparison Tables
Formulas & Exam Tips • Mnemonics • Revision Summaries
50 Revision Points • 60 Practice Questions • 20 MCQs • Glossary A-Z

Subject Area: Computer Applications / Data Science Fundamentals

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1. Introduction to Data Analytics

1.1 Definition

Data Analytics is the science of examining raw data with the aim of drawing conclusions, discovering patterns, and supporting decision-making. It combines statistics, computer science, and domain knowledge to turn data into useful information.

> **EXAM TIP**
> • A one-line definition is often asked in exams: "Data Analytics is the process of inspecting, cleaning, transforming, and modeling data to discover useful information and support decision-making."

1.2 Importance of Data Analytics

- Helps organizations make evidence-based decisions instead of guesswork.
- Improves efficiency by identifying bottlenecks and wastage in processes.
- Enables better understanding of customer behaviour and market trends.
- Reduces business risk by predicting future outcomes.
- Provides a competitive advantage in fast-changing markets.
- Supports automation and real-time decision-making in modern systems.

1.3 Applications of Data Analytics

Domain	Application Example
Business	Sales forecasting, market basket analysis
Healthcare	Disease prediction, patient monitoring
Banking & Finance	Fraud detection, credit scoring
E-commerce	Recommendation engines (Amazon, Flipkart)
Sports	Player performance analysis
Government	Census analysis, policy planning
Social Media	Sentiment analysis, trend detection

1.4 Types of Analytics

There are four major types of data analytics, based on the question they answer:

```
DESCRIPTIVE   ---->   DIAGNOSTIC   ---->   PREDICTIVE   ---->   PRESCRIPTIVE
What happened?        Why did it          What will             What should
                      happen?             happen?               we do?
(Past)                (Past - Cause)      (Future)              (Future - Action)
```

Type	Question Answered	Technique Used	Example
Descriptive	What happened?	Summary stats, dashboards	Monthly sales report
Diagnostic	Why did it happen?	Drill-down, correlation	Why sales dropped in June
Predictive	What will happen?	Regression, ML models	Forecasting next quarter sales

Type	Question Answered	Technique Used	Example
Prescriptive	What should we do?	Optimization, simulation	Suggesting best price/stock level

MEMORY TRICK (Mnemonic)

• Remember the order using "D-D-P-P": "Dead Data Predicts Profit" → Descriptive, Diagnostic, Predictive, Prescriptive (increasing complexity and value).

QUICK REVISION SUMMARY

• Data Analytics = examining data to find patterns and support decisions.
• Important because it enables data-driven decisions, reduces risk, improves efficiency.
• Applications span business, healthcare, finance, e-commerce, sports, government.
• 4 Types: Descriptive (past), Diagnostic (why), Predictive (future), Prescriptive (action).

2. Data Collection

2.1 Sources of Data

Data can be collected from a wide variety of sources:

- Surveys and questionnaires
- Interviews and observations
- Sensors and IoT devices
- Company databases and transaction records
- Web scraping and social media platforms
- Government reports and open data portals

2.2 Structured vs Unstructured Data

Basis	Structured Data	Unstructured Data
Format	Organized in rows & columns	No fixed format
Storage	Relational databases (SQL)	NoSQL, file systems, data lakes
Example	Excel sheet, bank records	Images, videos, emails, social posts
Ease of Analysis	Easy to search & analyze	Difficult, needs special tools (NLP, CV)
Approx. Volume Today	~20% of enterprise data	~80% of enterprise data

Semi-structured data is a middle category — it does not fit a rigid table but has some organizational tags/markers, e.g., XML files, JSON files, and emails with headers.

2.3 Primary vs Secondary Data

Basis	Primary Data	Secondary Data
Meaning	Collected first-hand for a specific purpose	Already collected by someone else, reused
Cost	Expensive and time-consuming	Cheaper and faster to obtain
Accuracy	More accurate & relevant to current need	May not perfectly match current need
Example	A company's own customer survey	Government census data, published reports

MEMORY TRICK (Mnemonic)

• "Primary = Personally collected", "Secondary = Someone else's data".

2.4 Data Quality

Good data must have the following characteristics, often remembered as the "5 Cs" of data quality:

Dimension	Meaning
Accuracy	Data correctly represents real-world values
Completeness	No important values are missing

Dimension	Meaning
Consistency	Same data does not contradict itself across sources
Timeliness	Data is up to date and available when needed
Validity	Data conforms to defined formats/rules
Uniqueness	No unwanted duplicate records

EXAM TIP

• Frequently asked: "What are the dimensions of data quality?" — Answer with Accuracy, Completeness, Consistency, Timeliness, Validity, Uniqueness (pick any 4-5 with one line each).

QUICK REVISION SUMMARY

• Data sources: surveys, sensors, databases, web scraping, government reports.
• Structured data = rows/columns (SQL); Unstructured = no fixed format (images, text).
• Primary data = collected first-hand; Secondary data = already existing data reused.
• Data quality dimensions: Accuracy, Completeness, Consistency, Timeliness, Validity, Uniqueness.

3. Data Preprocessing

Raw data collected from the real world is almost always messy. **Data preprocessing** prepares this raw data for analysis by cleaning and transforming it.

3.1 Data Cleaning

Data cleaning is the process of detecting and correcting (or removing) inaccurate, incomplete, or irrelevant parts of data.

- Removing duplicate records
- Fixing incorrect data types (e.g., text in a number column)
- Correcting spelling/format inconsistencies (e.g., 'Delhi' vs 'delhi')
- Removing irrelevant columns not needed for analysis

3.2 Handling Missing Values

Missing values occur when no data value is stored for a variable. Common strategies to handle them:

Method	Description	When to Use
Deletion	Remove rows/columns with missing values	When missing data is very small in amount
Mean/Median Imputation	Replace missing value with mean or median of column	For numeric data with few outliers
Mode Imputation	Replace with most frequent value	For categorical data
Forward/Backward Fill	Use previous/next value in sequence	For time-series data
Prediction Model	Use ML model to predict the missing value	When missing data is large & important

3.3 Outlier Detection

Outliers are data points that differ significantly from other observations and may indicate errors or rare events.

Common Methods to Detect Outliers

- **Box Plot Method:** Values outside 1.5 × IQR (Interquartile Range) from Q1 or Q3 are outliers.
- **Z-Score Method:** Values with $|Z| > 3$ are usually considered outliers.
- **Scatter Plot:** Visually identify points that lie far from the main cluster.

Formula: Z-Score

- $Z = (X - \mu) / \sigma$
- where X = data value, μ = mean, σ = standard deviation

Formula: IQR Rule for Outliers

- IQR = Q3 − Q1
- Lower Bound = Q1 − 1.5 × IQR
- Upper Bound = Q3 + 1.5 × IQR
- Any value outside [Lower Bound, Upper Bound] is an outlier.

3.4 Data Transformation

Transformation converts data into a suitable format or structure for analysis.

- Aggregation: Combining data (e.g., daily sales → monthly sales)
- Generalization: Converting low-level data into higher-level concepts (e.g., age → age group)
- Encoding: Converting categorical data into numeric form (e.g., Label Encoding, One-Hot Encoding)
- Discretization: Converting continuous data into intervals/bins

3.5 Normalization

Normalization rescales numeric data into a common range so that no variable dominates due to scale differences. This is essential before applying many ML algorithms.

Formula: Min-Max Normalization
• X' = (X − Xmin) / (Xmax − Xmin)
• Rescales data into range [0, 1]

Formula: Z-Score Standardization
• X' = (X − μ) / σ
• Rescales data to have mean = 0 and standard deviation = 1

EXAM TIP
• Do not confuse Normalization (rescaling to a range like 0-1) with Standardization (rescaling to mean 0, SD 1). Both may be asked to differentiate.

MEMORY TRICK (Mnemonic)
• "CMOTN" — Clean → Missing values → Outliers → Transform → Normalize: the typical order of preprocessing steps.

QUICK REVISION SUMMARY
• Data cleaning removes duplicates, errors, and irrelevant data.
• Missing values handled by deletion, mean/median/mode imputation, or prediction.
• Outliers detected using Box Plot (IQR rule) or Z-score method.
• Transformation includes aggregation, generalization, encoding, discretization.
• Normalization (Min-Max) rescales to [0,1]; Standardization (Z-score) gives mean 0, SD 1.

4. Exploratory Data Analysis (EDA)

EDA is the approach of analyzing datasets to summarize their main characteristics, often using visual methods, before formal modeling begins. It was popularized by statistician John Tukey.

4.1 Objectives of EDA

- Understand the structure and distribution of data
- Detect errors, missing values, and outliers
- Identify relationships and patterns between variables
- Test underlying assumptions before applying statistical models
- Select important features (variables) for modeling

4.2 Summary Statistics

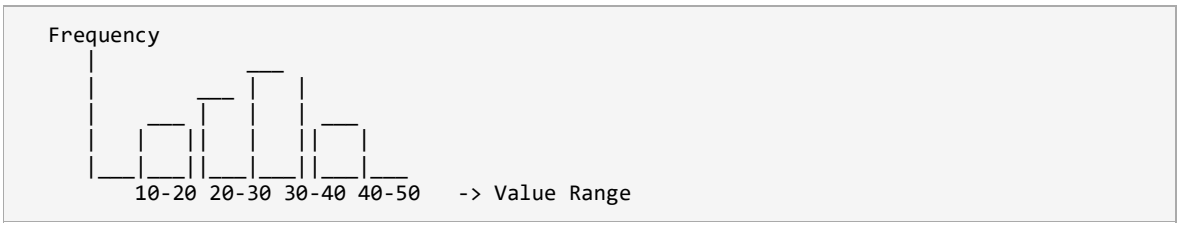
Summary statistics give a quick numeric overview of data, including measures of central tendency (mean, median, mode) and measures of spread (range, variance, standard deviation).

Measure	What It Tells Us
Mean	The average value of the dataset
Median	The middle value when data is sorted
Mode	The most frequently occurring value
Range	Difference between maximum and minimum
Variance/Std. Dev.	How spread out the data is from the mean

4.3 Data Visualization Techniques Used in EDA

Histogram

A histogram shows the frequency distribution of a single numeric variable using adjacent bars (bins).



Use: To see the shape of distribution — normal, skewed, bimodal, etc.

Scatter Plot

A scatter plot shows the relationship between two numeric variables using dots on an X-Y plane.



Use: To detect correlation, clusters, and outliers between two variables.

Box Plot (Box-and-Whisker Plot)

A box plot displays the 5-number summary: Minimum, Q1, Median, Q3, Maximum — and highlights outliers.



Use: To compare distributions and quickly detect outliers and spread.

4.4 Correlation

Correlation measures the strength and direction of a linear relationship between two numeric variables. It ranges from -1 to +1.

Correlation Coefficient (r) — Interpretation

- r = +1 → Perfect positive correlation
- r = 0 → No linear correlation
- r = −1 → Perfect negative correlation
- 0.7 to 1.0 (or −0.7 to −1.0) → Strong correlation

EXAM TIP

- Remember: Correlation does NOT imply causation! Two variables can be correlated without one causing the other (classic exam question).

MEMORY TRICK (Mnemonic)

- "HSB-C" for EDA visual tools: Histogram, Scatter plot, Box plot, Correlation.

QUICK REVISION SUMMARY

- EDA = analyzing data (mainly visually) before formal modeling.
- Objectives: understand structure, detect errors/outliers, find relationships.
- Key tools: Histogram (distribution), Scatter Plot (relationship), Box Plot (spread/outliers).
- Correlation (r) ranges from -1 to +1; correlation ≠ causation.

5. Data Visualization

Data visualization is the graphical representation of data and information, making complex data easier to understand, compare, and act upon.

5.1 Principles of Good Visualization

- Clarity: The message should be immediately understandable
- Accuracy: Do not distort scale, axis, or proportions
- Simplicity: Avoid unnecessary decoration ('chart junk')
- Right Chart Choice: Match the chart type to the type of data
- Consistency: Use uniform colours, fonts, and scales across a report

5.2 Common Charts and Graphs

Chart Type	Best Used For
Bar Chart	Comparing categories (e.g., sales by region)
Line Chart	Showing trends over time
Pie Chart	Showing proportion of a whole (few categories only)
Histogram	Showing frequency distribution of numeric data
Scatter Plot	Showing relationship between two variables
Heat Map	Showing intensity/density across a matrix
Box Plot	Showing spread and outliers

5.3 Dashboards

A dashboard is a visual display of the most important information needed to achieve one or more objectives, consolidated on a single screen so it can be monitored at a glance.

- Combine multiple charts/KPIs (Key Performance Indicators) in one view
- Support real-time or near real-time monitoring
- Common tools: Power BI, Tableau, Excel Dashboards, Google Data Studio

5.4 Best Practices

- Know your audience before choosing chart complexity
- Use colour meaningfully (e.g., red for negative, green for positive)
- Label axes and provide titles/legends
- Avoid 3D charts and excessive pie slices — they distort perception
- Highlight the key insight, don't just display raw numbers

EXAM TIP

• A common exam question: "Differentiate between a chart and a dashboard." — A chart shows a single visual; a dashboard is a collection of multiple charts/KPIs for monitoring at a glance.

QUICK REVISION SUMMARY

- Data visualization = graphical representation of data for easy understanding.
- Good visualization principles: clarity, accuracy, simplicity, right chart choice.
- Common charts: Bar, Line, Pie, Histogram, Scatter, Heat Map, Box Plot.
- Dashboards combine multiple visuals/KPIs; tools include Power BI and Tableau.

6. Statistical Concepts

Statistics forms the mathematical backbone of data analytics. This section covers essential measures used constantly in analysis.

6.1 Measures of Central Tendency

Mean (Average)

Formula: Mean
- Mean ($\bar{x}$) = (Σx) / n
- where Σx = sum of all values, n = number of values

Example: Data: 10, 20, 30, 40 → Mean = (10+20+30+40)/4 = 25

Median

The median is the middle value of a data set arranged in ascending or descending order.

- If n is odd: Median = value at position (n+1)/2
- If n is even: Median = average of the two middle values

Example: Data: 3, 5, 7, 9, 11 → Median = 7 (middle value)

Mode

The mode is the value that occurs most frequently in a data set. A data set can have one mode, more than one mode (bimodal/multimodal), or no mode.

Example: Data: 2, 3, 3, 5, 7 → Mode = 3

MEMORY TRICK (Mnemonic)
- "Mean = Add & Divide, Median = Middle, Mode = Most" — a quick way to remember the three.

6.2 Measures of Dispersion

Variance

Formula: Variance (Population)
- $\sigma^2 = \Sigma(x - \mu)^2 / N$
- where μ = mean, N = total number of values

Standard Deviation

Formula: Standard Deviation
- $\sigma = \sqrt{\text{Variance}} = \sqrt{[\Sigma(x - \mu)^2 / N]}$

Standard deviation shows how much data values deviate from the mean, on average. A **low SD** means data points are close to the mean; a **high SD** means data is widely spread.

EXAM TIP
- Exam favourite: Calculate mean, variance, and SD for a small data set (5-6 numbers). Practice this calculation by hand.

6.3 Probability Basics

Probability measures the likelihood of an event occurring, expressed as a number between 0 (impossible) and 1 (certain).

Formula: Basic Probability

• P(Event) = (Number of favourable outcomes) / (Total number of outcomes)

Term	Meaning
Experiment	An action with an uncertain outcome (e.g., tossing a coin)
Sample Space	Set of all possible outcomes
Event	A subset of the sample space
Independent Events	Occurrence of one does not affect the other
Mutually Exclusive Events	Events that cannot occur at the same time

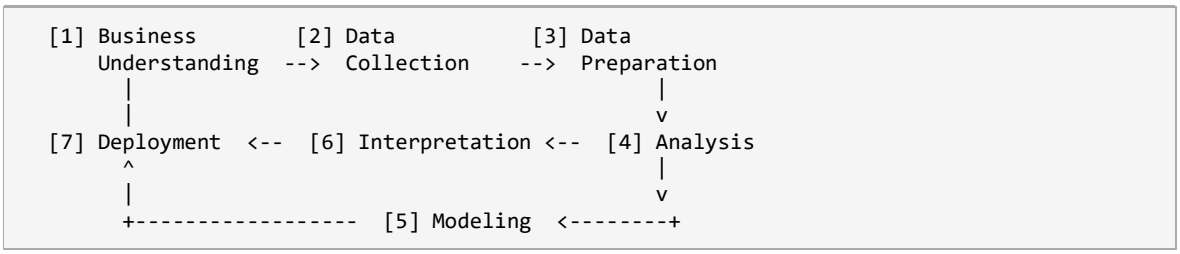
Example: Probability of rolling a 4 on a fair six-sided die = 1/6.

QUICK REVISION SUMMARY

• Mean = average; Median = middle value; Mode = most frequent value.
• Variance measures spread of data; Standard Deviation = √Variance.
• Low SD = data close to mean; High SD = data widely spread.
• Probability = favourable outcomes / total outcomes; ranges from 0 to 1.

7. Data Analytics Process

A structured process ensures that analytics projects deliver reliable and actionable results. The typical process has seven stages:



7.1 Business Understanding

Define the business problem, objectives, and success criteria. Example: "Reduce customer churn by 10% in 6 months."

7.2 Data Collection

Gather relevant data from internal databases, external sources, surveys, or APIs (as covered in Section 2).

7.3 Data Preparation

Clean, transform, and organize data into a usable format (as covered in Section 3 — Data Preprocessing).

7.4 Analysis

Apply exploratory and statistical techniques to uncover patterns, trends, and relationships in the data.

7.5 Modeling

Build statistical or machine learning models (regression, classification, clustering) to predict outcomes or classify data.

7.6 Interpretation

Translate model results and analysis findings into business insights that non-technical stakeholders can understand.

7.7 Deployment

Implement the solution into real business operations — e.g., integrating a fraud-detection model into a bank's transaction system — and monitor performance over time.

EXAM TIP

• This 7-step process is very similar to the industry-standard CRISP-DM (Cross Industry Standard Process for Data Mining) model — mentioning CRISP-DM by name can fetch bonus marks.

MEMORY TRICK (Mnemonic)

• "Big Data Projects Aim Massive Impact Daily" → Business Understanding, Data Collection, Preparation, Analysis, Modeling, Interpretation, Deployment.

QUICK REVISION SUMMARY

- 7 Stages: Business Understanding → Data Collection → Preparation → Analysis → Modeling → Interpretation → Deployment.
- Process is cyclic — insights from deployment can trigger new business questions.
- Related standard: CRISP-DM methodology.

8. Tools Used in Data Analytics

Different tools serve different purposes across the analytics process — from simple spreadsheets to advanced programming languages.

8.1 Microsoft Excel

- Spreadsheet tool for basic data storage, cleaning, and analysis
- Features: Pivot Tables, VLOOKUP/XLOOKUP, charts, basic statistical functions
- Best for: small datasets, quick analysis, business reporting

8.2 SQL (Structured Query Language)

- Used to query, filter, and manage data stored in relational databases
- Common commands: SELECT, WHERE, GROUP BY, JOIN, ORDER BY
- Best for: extracting and aggregating data from large structured databases

8.3 Python

- A general-purpose programming language, widely used in analytics and ML
- Key libraries: Pandas (data handling), NumPy (numeric computing), Matplotlib/Seaborn (visualization), Scikit-learn (machine learning)
- Best for: end-to-end analytics pipelines, automation, machine learning

8.4 R

- A programming language built specifically for statistics and data analysis
- Strong in statistical modeling and academic/research-based analytics
- Popular packages: ggplot2 (visualization), dplyr (data manipulation)

8.5 Tableau

- A leading data visualization and business intelligence (BI) tool
- Drag-and-drop interface to build interactive dashboards
- Best for: visually rich, interactive reports for business users

8.6 Power BI

- Microsoft's business intelligence tool for interactive visualizations and dashboards
- Integrates well with Excel and other Microsoft products
- Best for: enterprise reporting connected to Microsoft ecosystem

8.7 Comparison Table

Tool	Type	Skill Level	Primary Use
Excel	Spreadsheet	Beginner	Basic analysis & reporting
SQL	Query Language	Beginner-Intermediate	Data extraction from databases
Python	Programming Language	Intermediate-Advanced	Full analytics & ML pipeline
R	Programming Language	Intermediate-Advanced	Statistical analysis & research

Tool	Type	Skill Level	Primary Use
Tableau	BI/Visualization Tool	Beginner-Intermediate	Interactive dashboards
Power BI	BI/Visualization Tool	Beginner-Intermediate	Enterprise dashboards & reports

EXAM TIP

• A common question: "Compare Python and R for data analytics." Key point: Python is more general-purpose and better for deployment/ML; R is more specialized for statistics and academic research.

QUICK REVISION SUMMARY

• Excel: quick/basic analysis. SQL: querying databases.
• Python: full-stack analytics + ML (Pandas, NumPy, Scikit-learn).
• R: strong for statistics and research.
• Tableau & Power BI: leading tools for interactive dashboards & BI reporting.

9. Machine Learning in Data Analytics

Machine Learning (ML) allows systems to learn patterns from data and make predictions or decisions without being explicitly programmed for every rule.

9.1 Supervised Learning

The model is trained on **labeled data** (input paired with correct output). The goal is to learn a mapping from input to output.

- Used when historical outcomes are known
- Two main types: Regression and Classification

Regression

Predicts a **continuous numeric value**.

Example: Predicting house price based on area, location, and number of rooms.

> **Simple Linear Regression Equation**
> • Y = a + bX
> • where Y = predicted value, X = input variable, a = intercept, b = slope

Classification

Predicts a **category or class label**.

Example: Classifying an email as "Spam" or "Not Spam".

- Common algorithms: Logistic Regression, Decision Tree, Random Forest, Naive Bayes, Support Vector Machine (SVM)

9.2 Unsupervised Learning

The model is trained on **unlabeled data** — there is no predefined output. The system tries to find hidden patterns or groupings on its own.

Clustering

Groups similar data points together based on their characteristics.

Example: Grouping customers into segments based on purchasing behaviour (Customer Segmentation).

- Common algorithm: K-Means Clustering

9.3 Supervised vs Unsupervised — Comparison

Basis	Supervised Learning	Unsupervised Learning
Data Used	Labeled data (input + known output)	Unlabeled data (no known output)
Goal	Predict output for new data	Discover hidden patterns/groups
Examples	Regression, Classification	Clustering, Association
Use Case	Spam detection, price prediction	Customer segmentation, anomaly detection

```
        MACHINE LEARNING
        /              \
```

```
  SUPERVISED        UNSUPERVISED
   /      \               |
Regression Classification Clustering
(numeric)   (category)    (grouping)
```

MEMORY TRICK (Mnemonic)

• "Supervised = Student with Answer Key, Unsupervised = Student without Answer Key" — Supervised learning has known correct answers (labels); unsupervised does not.

EXAM TIP

• Classic exam question: "Differentiate between Classification and Regression." Key point — Regression predicts continuous numbers; Classification predicts discrete categories.

QUICK REVISION SUMMARY

• Supervised Learning uses labeled data → Regression (numeric output) & Classification (categorical output).
• Unsupervised Learning uses unlabeled data → Clustering (grouping similar data, e.g., K-Means).
• Regression example: predicting house price. Classification example: spam detection.
• Clustering example: customer segmentation.

10. Business Applications

Data analytics is applied across nearly every industry. Below are key applications in six major sectors.

10.1 Finance

- Fraud detection using anomaly detection algorithms
- Credit scoring and risk assessment for loan approvals
- Algorithmic trading based on predictive models
- Customer segmentation for personalized financial products

10.2 Healthcare

- Predicting disease outbreaks and patient risk factors
- Personalized treatment plans based on patient history
- Medical image analysis (X-rays, MRI scans) using ML
- Hospital resource management (bed allocation, staff scheduling)

10.3 Retail

- Market basket analysis ("Customers who bought X also bought Y")
- Inventory and demand forecasting
- Dynamic pricing based on demand and competitor prices
- Customer churn prediction and loyalty programs

10.4 Marketing

- Customer segmentation and targeted advertising
- Sentiment analysis of social media and reviews
- Campaign performance tracking (click-through rate, conversion rate)
- Recommendation systems (as used by Netflix, YouTube)

10.5 Manufacturing

- Predictive maintenance to reduce machine downtime
- Quality control using defect-detection analytics
- Supply chain optimization
- Production planning based on demand forecasts

10.6 Education

- Student performance prediction and early dropout warnings
- Personalized learning paths based on student data
- Analyzing course effectiveness through engagement data
- Resource allocation planning for institutions

EXAM TIP

• For "Business Applications" questions, always give ONE concrete example per domain rather than generic statements — examiners reward specificity.

QUICK REVISION SUMMARY

- Finance: fraud detection, credit scoring. Healthcare: disease prediction, treatment plans.
- Retail: market basket analysis, demand forecasting. Marketing: segmentation, recommendations.
- Manufacturing: predictive maintenance, quality control. Education: dropout prediction, personalized learning.

11. Challenges in Data Analytics

11.1 Privacy

Collecting and analyzing personal data raises concerns about how that data is used, stored, and shared. Laws like GDPR (Europe) and India's Digital Personal Data Protection Act regulate this.

11.2 Security

Large data repositories are attractive targets for cyberattacks. Organizations must protect data through encryption, access control, and regular security audits.

11.3 Data Quality

Poor quality data (incomplete, inconsistent, outdated) leads to unreliable analysis and wrong business decisions — often summarized as "Garbage In, Garbage Out (GIGO)."

11.4 Ethical Issues

- Using data without proper consent
- Discriminatory targeting based on sensitive attributes
- Lack of transparency in how algorithms make decisions ('black box' models)

11.5 Bias

Bias occurs when data or algorithms systematically favour or disadvantage certain groups, often because the training data itself is not representative.

Type of Bias	Description
Sampling Bias	Training data does not represent the real population
Confirmation Bias	Analysts favour data that confirms existing beliefs
Algorithmic Bias	Model produces systematically unfair outcomes for certain groups

EXAM TIP

• GIGO (Garbage In, Garbage Out) is a very popular exam term — always mention it under Data Quality challenges.

MEMORY TRICK (Mnemonic)

• "PS-DEB" — Privacy, Security, Data Quality, Ethical issues, Bias: the five key challenges.

QUICK REVISION SUMMARY

• Privacy: concerns over personal data usage (GDPR, DPDP Act).
• Security: protecting data from breaches/cyberattacks.
• Data Quality: GIGO — poor data leads to poor decisions.
• Ethical Issues: consent, discrimination, lack of transparency.
• Bias: sampling, confirmation, and algorithmic bias distort fair outcomes.

12. Future Trends

12.1 AI-driven Analytics

Artificial Intelligence is increasingly automating the analytics process itself — from automatically cleaning data to selecting the best model (AutoML).

12.2 Big Data

Big Data refers to extremely large and complex datasets that traditional tools cannot process efficiently, commonly described using the **5 Vs**:

V	Meaning
Volume	Huge amount of data generated
Velocity	Speed at which data is generated and processed
Variety	Different types/formats of data (text, image, video)
Veracity	Trustworthiness/quality of data
Value	Usefulness of data for decision-making

MEMORY TRICK (Mnemonic)

- "5 Vs of Big Data": Volume, Velocity, Variety, Veracity, Value.

12.3 Cloud Analytics

Performing data storage and analytics on cloud platforms (AWS, Microsoft Azure, Google Cloud) rather than local servers, offering scalability and lower infrastructure cost.

12.4 Real-Time Analytics

Analyzing data as it is generated to enable immediate decision-making — e.g., fraud alerts during a transaction, live traffic navigation updates.

12.5 Generative AI

Generative AI (like large language models) is being combined with analytics to auto-generate insights, reports, and even natural-language explanations of complex data trends, making analytics accessible to non-technical users.

EXAM TIP

- A frequently asked question: "What are the 5 Vs of Big Data?" — make sure to list and briefly explain all five.

QUICK REVISION SUMMARY

- AI-driven analytics automates cleaning, modeling, and insight generation (AutoML).
- Big Data described by 5 Vs: Volume, Velocity, Variety, Veracity, Value.
- Cloud Analytics uses platforms like AWS/Azure/GCP for scalable processing.
- Real-Time Analytics enables instant decisions (fraud alerts, live tracking).
- Generative AI is enabling automatic, natural-language insight generation.

13. 50 Important One-Line Revision Points

Use this list for last-minute revision before the exam. Each point is a self-contained, exam-ready statement.

1. Data Analytics is the process of examining data to draw conclusions and support decisions.

2. Four types of analytics: Descriptive, Diagnostic, Predictive, Prescriptive.

3. Descriptive analytics answers 'What happened?'; Prescriptive answers 'What should we do?'

4. Structured data fits rows/columns; unstructured data has no fixed format.

5. Primary data is collected first-hand; secondary data is already existing.

6. Data quality dimensions: Accuracy, Completeness, Consistency, Timeliness, Validity, Uniqueness.

7. Data cleaning removes duplicates, errors, and irrelevant records.

8. Missing values can be handled by deletion, mean/median/mode imputation, or prediction models.

9. Outliers are detected using the IQR (Box Plot) rule or Z-score method.

10. IQR = Q3 − Q1; outlier bounds = Q1 − 1.5×IQR and Q3 + 1.5×IQR.

11. Normalization (Min-Max) rescales data to range [0,1].

12. Standardization (Z-score) rescales data to mean 0 and standard deviation 1.

13. EDA (Exploratory Data Analysis) summarizes data characteristics, often visually, before modeling.

14. Histogram shows frequency distribution of a single numeric variable.

15. Scatter plot shows the relationship between two numeric variables.

16. Box plot shows the 5-number summary and highlights outliers.

17. Correlation coefficient (r) ranges from -1 to +1; correlation does not imply causation.

18. Good visualization principles: clarity, accuracy, simplicity, correct chart choice.

19. A dashboard combines multiple charts/KPIs into a single monitoring view.

20. Mean = sum of values ÷ number of values.

21. Median is the middle value of sorted data.

22. Mode is the most frequently occurring value.

23. Variance measures the average squared deviation from the mean.

24. Standard Deviation = square root of Variance.

25. Probability = favourable outcomes ÷ total outcomes; ranges from 0 to 1.

26. The Data Analytics Process has 7 stages: Business Understanding, Data Collection, Preparation, Analysis, Modeling, Interpretation, Deployment.

27. CRISP-DM is the industry-standard methodology similar to the analytics process.

28. Excel is best for small datasets and quick business reporting.

29. SQL is used to query and manage data in relational databases.

30. Python is widely used for full analytics pipelines and machine learning (Pandas, NumPy, Scikit-learn).

31. R is specialized for statistical analysis and academic research.

32. Tableau and Power BI are leading business intelligence (BI) and dashboard tools.

33. Supervised learning uses labeled data; unsupervised learning uses unlabeled data.

34. Regression predicts a continuous numeric value (e.g., house price).

35. Classification predicts a discrete category (e.g., spam or not spam).

36. Clustering groups similar data points together (e.g., K-Means for customer segmentation).

37. Linear Regression equation: Y = a + bX.

38. Data Analytics is applied in Finance (fraud detection), Healthcare (disease prediction), Retail (demand forecasting).

39. Marketing uses analytics for segmentation, sentiment analysis, and recommendation systems.

40. Manufacturing uses predictive maintenance and quality control analytics.

41. Education uses analytics for dropout prediction and personalized learning.

42. Key challenges in analytics: Privacy, Security, Data Quality, Ethical Issues, Bias.

43. GIGO (Garbage In, Garbage Out) means poor input data leads to poor analysis results.

44. Bias types include Sampling Bias, Confirmation Bias, and Algorithmic Bias.

45. Big Data is described using the 5 Vs: Volume, Velocity, Variety, Veracity, Value.

46. Cloud Analytics uses platforms like AWS, Azure, and Google Cloud for scalable data processing.

47. Real-Time Analytics processes data instantly for immediate decisions (e.g., fraud alerts).

48. Generative AI is being used to auto-generate insights and natural-language explanations of data.

49. AI-driven analytics (AutoML) automates data cleaning, model selection, and tuning.

50. Data Analytics combines statistics, computer science, and domain knowledge to create value from data.

14. 25 University-Style Short-Answer Questions

Instructions: Answer each question in 2-5 sentences. These are typically worth 2-5 marks each.

1. Define Data Analytics.

2. List the four types of analytics with one example each.

3. Differentiate between structured and unstructured data.

4. What is the difference between primary and secondary data?

5. List any four dimensions of data quality.

6. What is data cleaning? Give two examples of cleaning operations.

7. How can missing values be handled in a dataset?

8. What is an outlier? Name two methods to detect it.

9. Write the formula for the Z-score and explain its terms.

10. Differentiate between Normalization and Standardization.

11. What are the objectives of Exploratory Data Analysis (EDA)?

12. What is a box plot used for?

13. Define correlation. What does a correlation of -1 mean?

14. State any three principles of good data visualization.

15. What is a dashboard? Name two tools used to build dashboards.

16. Define Mean, Median, and Mode with one example each.

17. What is Standard Deviation and what does it indicate?

18. Define probability with a suitable example.

19. List the seven stages of the Data Analytics Process.

20. Compare Python and R as data analytics tools.

21. Differentiate between Supervised and Unsupervised Learning.

22. What is the difference between Regression and Classification?

23. Give one real-world application of data analytics in Healthcare and Retail.

24. What is meant by 'Bias' in data analytics? Name its types.

25. Explain the 5 Vs of Big Data.

15. 15 Long-Answer Questions

Instructions: Answer each question in detail with examples, diagrams, and tables wherever applicable. These are typically worth 8-15 marks each.

1. Explain the four types of data analytics (Descriptive, Diagnostic, Predictive, Prescriptive) with suitable real-world examples for each.

2. Discuss various sources of data collection and explain the differences between structured, unstructured, and semi-structured data.

3. What is data preprocessing? Explain the steps involved in preparing raw data for analysis with examples.

4. Explain the concept of missing values and outliers. Describe methods used to handle each with suitable examples.

5. What is Exploratory Data Analysis? Explain the key visualization techniques (Histogram, Scatter Plot, Box Plot) used in EDA with diagrams.

6. Explain the principles of good data visualization and describe the role of dashboards in business decision-making.

7. Explain measures of central tendency and dispersion with formulas, including Mean, Median, Mode, Variance, and Standard Deviation.

8. Describe the complete Data Analytics Process from Business Understanding to Deployment with a flowchart.

9. Compare and contrast the major tools used in Data Analytics: Excel, SQL, Python, R, Tableau, and Power BI.

10. Explain Supervised and Unsupervised Machine Learning with their types (Regression, Classification, Clustering) and real-world examples.

11. Discuss the applications of Data Analytics in any four business domains of your choice with examples.

12. Explain the major challenges faced in Data Analytics including privacy, security, data quality, ethics, and bias.

13. Discuss emerging trends in Data Analytics including AI-driven analytics, Big Data, Cloud Analytics, and Generative AI.

14. Explain the importance of data quality and describe the consequences of using poor-quality data in analytics (GIGO concept).

15. With the help of a diagram, explain how correlation and regression are used to understand relationships between variables.

16. 20 MCQs with Answers

Instructions: Choose the correct option. Answer key is provided at the end.

1. Which type of analytics answers the question 'Why did it happen?'

A) Descriptive
B) Diagnostic
C) Predictive
D) Prescriptive

2. Which of the following is an example of unstructured data?

A) Excel sheet
B) SQL table
C) Video file
D) Bank ledger

3. Data collected first-hand for a specific research purpose is called:

A) Secondary data
B) Primary data
C) Meta data
D) Big data

4. Which is NOT a dimension of data quality?

A) Accuracy
B) Timeliness
C) Popularity
D) Consistency

5. Replacing a missing numeric value with the column's average is called:

A) Deletion
B) Mean Imputation
C) Normalization
D) Encoding

6. The formula $Z = (X - \mu)/\sigma$ is used to calculate:

A) Mean
B) Median
C) Z-score
D) Mode

7. Min-Max Normalization rescales data into the range:

A) [-1, 1]
B) [0, 1]
C) [0, 100]
D) [1, 10]

8. Which chart is best suited to show the relationship between two numeric variables?

A) Pie Chart
B) Histogram
C) Scatter Plot
D) Bar Chart

9. A correlation coefficient of 0 indicates:

A) Strong positive relation
B) Strong negative relation
C) No linear relationship
D) Perfect relationship

10. Which measure of central tendency is most affected by outliers?

A) Median
B) Mode
C) Mean
D) Range

11. Standard Deviation is calculated as:

A) Variance squared
B) Square root of Variance
C) Mean of Variance
D) Variance divided by n

12. Which is the first stage of the Data Analytics Process?

A) Data Collection
B) Modeling
C) Business Understanding
D) Deployment

13. Which tool is best known for statistical research and the ggplot2 package?

A) Excel
B) R
C) Power BI
D) SQL

14. SQL is primarily used for:

A) Data visualization
B) Querying relational databases
C) Machine learning modeling
D) Statistical testing

15. In Machine Learning, predicting house prices is an example of:

A) Classification
B) Clustering
C) Regression
D) Association

16. K-Means is a popular algorithm for:

A) Regression
B) Classification
C) Clustering
D) Normalization

17. Which learning type uses labeled data?

A) Unsupervised Learning
B) Reinforcement Learning
C) Supervised Learning
D) Deep Clustering

18. GIGO stands for:

A) Good In, Good Out
B) Garbage In, Garbage Out
C) Grouped Input, Grouped Output
D) Generic In, Generic Out

19. Which of the 5 Vs of Big Data refers to the trustworthiness of data?

A) Volume
B) Velocity
C) Veracity

D) Variety

20. Which of the following is an example of Prescriptive Analytics?

A) Monthly sales report
B) Predicting next month's sales
C) Recommending optimal stock levels
D) Explaining why sales dropped

Answer Key

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	6	C	11	B	16	C
2	C	7	B	12	C	17	C
3	B	8	C	13	B	18	B
4	C	9	C	14	B	19	C
5	B	10	C	15	C	20	C

17. Glossary of Important Data Analytics Terms (A–Z)

Term	Definition
Algorithm	A step-by-step procedure or set of rules used to solve a problem or perform a computation.
Big Data	Extremely large and complex datasets characterized by Volume, Velocity, Variety, Veracity, and Value.
Classification	A supervised ML technique that predicts a discrete category or label.
Clustering	An unsupervised ML technique that groups similar data points together.
Correlation	A statistical measure indicating the strength and direction of a linear relationship between two variables.
Dashboard	A visual display consolidating key metrics/charts for monitoring at a glance.
Data Cleaning	The process of detecting and correcting inaccurate or irrelevant data.
Data Mining	The process of discovering patterns and knowledge from large datasets.
Data Preprocessing	Preparing raw data (cleaning, transforming) before analysis.
Descriptive Analytics	Analytics that summarizes what has happened in the past.
EDA	Exploratory Data Analysis — analyzing data sets to summarize their main characteristics, often visually.
Feature	An individual measurable property or variable used as input to a model.
GIGO	Garbage In, Garbage Out — poor quality input data leads to poor quality output/results.
Histogram	A chart showing the frequency distribution of a numeric variable using bars.
Imputation	The process of replacing missing data with substituted values.
Interquartile Range (IQR)	The range between the first quartile (Q1) and third quartile (Q3); used to detect outliers.
K-Means	A popular unsupervised clustering algorithm that groups data into K clusters.
Machine Learning (ML)	A branch of AI where systems learn patterns from data to make predictions/decisions.
Mean	The arithmetic average of a set of numbers.
Median	The middle value in a sorted dataset.
Mode	The most frequently occurring value in a dataset.
Normalization	Rescaling numeric data into a common range, typically [0,1].
Outlier	A data point significantly different from other observations in a dataset.
Predictive Analytics	Analytics that forecasts future outcomes using historical data.
Prescriptive Analytics	Analytics that recommends actions to achieve desired outcomes.
Primary Data	Data collected first-hand for a specific purpose.
Probability	A measure of the likelihood that an event will occur, from 0 to 1.
Regression	A supervised ML technique that predicts a continuous numeric value.
Secondary Data	Data that already exists, collected earlier by someone else.

Term	Definition
Standard Deviation	A measure of how spread out data values are from the mean.
Structured Data	Data organized in a fixed format such as rows and columns.
Supervised Learning	ML approach that learns from labeled data (input-output pairs).
Unstructured Data	Data with no predefined format, such as text, images, or videos.
Unsupervised Learning	ML approach that finds patterns in data without labeled outputs.
Variance	The average of the squared differences between each data point and the mean.
Visualization	The graphical representation of data to communicate information clearly.

18. Quick Revision Sheet

A condensed 2-page recap of the entire subject — ideal for the night before the exam.

A. Core Definitions

Concept	One-Line Definition
Data Analytics	Examining data to discover patterns and support decisions
Descriptive Analytics	What happened? (past summary)
Diagnostic Analytics	Why did it happen? (root cause)
Predictive Analytics	What will happen? (forecast)
Prescriptive Analytics	What should we do? (recommendation)
EDA	Analyzing data visually/statistically before modeling
Normalization	Rescale data to [0,1]
Standardization	Rescale data to mean 0, SD 1

B. Key Formulas at a Glance

All Essential Formulas

- Mean ($\bar{x}$) = $\Sigma x / n$
- Variance (σ^2) = $\Sigma(x - \mu)^2 / N$
- Standard Deviation (σ) = $\sqrt{\text{Variance}}$
- Z-score = $(X - \mu) / \sigma$
- IQR = Q3 − Q1 | Outlier bounds: Q1 − 1.5×IQR, Q3 + 1.5×IQR
- Min-Max Normalization: X' = (X − Xmin) / (Xmax − Xmin)
- Linear Regression: Y = a + bX
- Probability: P(Event) = Favourable Outcomes / Total Outcomes

C. Data Types Cheat-Sheet

Category	Key Point
Structured Data	Rows & columns, stored in SQL databases
Unstructured Data	No fixed format — text, images, video
Primary Data	First-hand, collected specifically for the study
Secondary Data	Already existing, reused from other sources

D. EDA Visual Tools

- Histogram → distribution of ONE variable
- Scatter Plot → relationship between TWO variables
- Box Plot → spread + outliers (5-number summary)
- Correlation (r) → -1 to +1; does NOT imply causation

E. Data Analytics Process (7 Stages)

Business Understanding → Data Collection → Data Preparation → Analysis → Modeling → Interpretation → Deployment

F. Tools Summary

Tool	One-Line Use
Excel	Basic spreadsheet analysis & reporting
SQL	Querying relational databases
Python	Full analytics + ML pipeline (Pandas, Scikit-learn)
R	Statistical analysis & research
Tableau / Power BI	Interactive dashboards & business intelligence

G. Machine Learning Snapshot

Category	Sub-type	Predicts	Example
Supervised	Regression	Continuous number	House price prediction
Supervised	Classification	Category/class	Spam email detection
Unsupervised	Clustering	Groups/segments	Customer segmentation

H. Challenges & Trends (Fast Recall)

- Challenges: Privacy, Security, Data Quality (GIGO), Ethical Issues, Bias
- Big Data = 5 Vs → Volume, Velocity, Variety, Veracity, Value
- Future Trends → AI-driven Analytics, Big Data, Cloud Analytics, Real-Time Analytics, Generative AI

I. Last-Minute Mnemonics

MEMORY TRICK (Mnemonic)
• Types of Analytics — "Dead Data Predicts Profit" (Descriptive, Diagnostic, Predictive, Prescriptive)

MEMORY TRICK (Mnemonic)
• Preprocessing Order — "CMOTN" (Clean, Missing values, Outliers, Transform, Normalize)

MEMORY TRICK (Mnemonic)
• Big Data — "5 Vs" (Volume, Velocity, Variety, Veracity, Value)

MEMORY TRICK (Mnemonic)
• EDA Tools — "HSB-C" (Histogram, Scatter plot, Box plot, Correlation)

EXAM TIP
• Before entering the exam hall, re-read Section A (Definitions), Section B (Formulas), and the Mnemonics one final time — these cover the majority of 1-5 mark questions.

— End of Study Notes — All the best for your examination! —